

Prompt Probes Read Solve Rate, Not Robustness: Negative Results from the AIMO Interpretability Challenge (Small Models Track)

Haeyoung Park
Sunrin Internet High School, Seoul, Korea

Abstract. We tried to predict, from a single forward pass over the problem prompt, whether a small reasoning model solves an olympiad problem robustly. A linear probe on one model's prompt representation looked strong in cross-validation on the official training rows (balanced accuracy up to 0.78), but those rows never assign different labels to different models on the same problem, so the probe could succeed by reading problem difficulty. To test the property the competition actually scores, we generated our own robustness labels for 150 new problems and three of the four small-track models (8 samples on the original, 18 on rule-based perturbations, an 8k-token reasoning budget with forced answers). Three findings follow. (1) A model's own hidden state predicts its own solve rate well (out-of-fold $r = 0.69$ to 0.75), better than the answer agreement of four sampled solutions ($r = 0.53$ on held-out samples). (2) It does not predict robustness within a model: once model identity is removed, self-probes are at chance (0.42 to 0.49) on problems balanced per label, and their apparent gains (0.63 vs 0.54 majority) come from model-identity priors that the official validation set is built to neutralize. (3) A cheap test-time counterfactual check (does the model keep its own majority answer under rename, typo and distractor variants?) reaches 0.64 to 0.66 against our own labels but 0.38 (balanced 0.48) against the organizers' labels and 0.50 on the leaderboard: agreement came from sharing our labeling choices. Our final entry scores 0.643 on the 14-case validation set. We document two pitfalls that silently corrupt self-generated labels (reasoning truncation and an fp8 KV-cache failure) and release all code and labels.

1. Task and setting

For a model m and an olympiad problem q , the task is to predict whether m 's accuracy on q holds up under perturbations that keep the deductive chain (robust) or drops clearly under at least one (spurious) [1]. We entered the Small Models track: Qwen3.5-4B, Skywork-OR1-Math-7B, Olmo-3-7B-Think and DeepSeek-R1-0528-Qwen3-8B. Submissions run offline on one RTX PRO 6000 with 3600 s for the whole prediction run. Public supervision is thin: 34 labeled rows over 20 problems for the four small models. The organizers state that the validation and test sets are balanced per problem and per model, so that neither model identity nor problem features alone are predictive; an always-true submission scores exactly 0.50 on the 14-case small-track validation set.

2. A shared prompt probe reads problem difficulty

We extracted hidden states of every layer for the final prompt token and the mean over prompt tokens, for every labeled problem under every model, and fitted L2 logistic probes with problem-grouped 5-fold cross-validation (3 seeds). A *shared* probe uses one encoder's hidden state for every target model plus a one-hot of the target; a *self* probe uses the target model's own hidden state. Figure 1 shows balanced accuracy on the 34 official rows by relative depth.

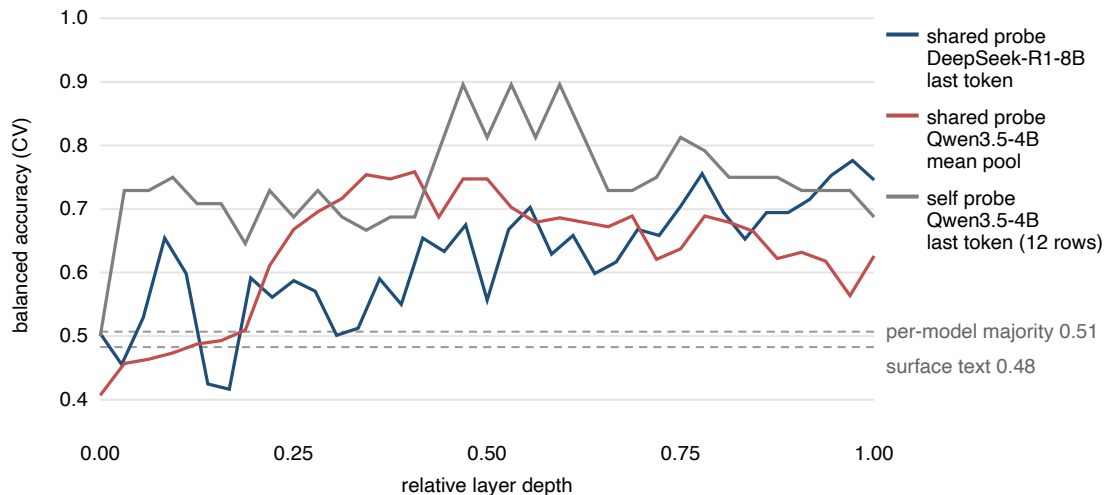


Figure 1. Out-of-fold balanced accuracy on the 34 official small-track rows. The shared DeepSeek-R1-8B last-token probe rises with depth (mean 0.72 over the top quarter, best layer 0.78). Surface text features (length, digits, LaTeX density) and the per-model majority stay at chance.

The curve looks like a real, depth-localized signal. It is not evidence for the competition's target. Of the 9 problems that carry labels for more than one small model, none has different labels across models, so a probe that only reads "this problem is hard" can score well here. The shared probe adds the target only as a bias term, so it cannot assign different labels to different models on the same problem. Submitted (layer 34, trained on all public rows), it scored 0.643 (9/14) on the validation set, two cases above always-true.

3. Self-generated labels

To evaluate on what is scored, we labeled 150 integer-answer problems not in the official data (AIME 2022 to 2026, AMC 12 2022 to 2023, and smaller sets from HMMT, BRUMO, SMT and CMIMC 2025 and 2026) for three models, using the organizers' sampling settings ($T = 1.0$, top-k 40, top-p 0.95) with vLLM. Each model answers the original 8 times; problems it solves at least once get one variant per family (rename, typos, distract; rule-based) answered 6 times each. With base accuracy b and family accuracy a_f , the label is robust if $\max_i(b - a_f) \leq 0.10$, spurious if ≥ 0.25 , and unlabeled in between, following the public training-set definition. A model that never solves a problem is labeled robust, since its accuracy cannot drop. Olmo-3-7B-Think was dropped: its full multi-head KV cache allowed only about 5 concurrent sequences on a 32 GB GPU (640 tokens/s), more than four hours for the first stage alone.

Two pitfalls that silently corrupt labels

- **Truncation.** At a 12k-token cap, 90% of Qwen3.5-4B samples were cut off before an answer and its mean base accuracy was 0.10, turning "ran out of budget" into "cannot solve". We cap reasoning at 8k tokens and force an answer by closing the reasoning and decoding `\boxed{ greedily` (the thinking-budget idea in the organizers' FAQ). Mean base accuracy became 0.49 (Qwen), 0.50 (Skywork) and 0.28 (DeepSeek), with 3% to 8% unparseable answers. Our labels therefore describe robustness at a fixed 8k reasoning budget.
- **fp8 KV cache.** With an fp8 KV cache in vLLM 0.31, Skywork-OR1-Math-7B (Qwen2.5 base) degenerated into single-token loops ("2222...", "the the the..."); accuracy of non-truncated answers was 0.12. With a bf16 cache it was 0.98. Qwen3.5-4B and DeepSeek-R1-8B were unaffected (non-truncated accuracy 1.00).

The result is 393 labeled rows. 44 of the 150 problems have different labels across models (117 rows); these are the rows that resemble the per-problem balanced test set.

4. Self-probes: solve rate yes, robustness no

Table 1 reports out-of-fold accuracy (problem-grouped folds, 3 seeds) on the mixed problems and on the 25 official rows of the same three models. The best configuration blends two routes on the target's own hidden state (mean-pooled, layers at relative depth 0.5 to 0.7): a ridge regression to the model's own solve rate followed by a logistic map to robustness, and a direct logistic probe.

Method	mixed problems (117 rows)	official rows (25)
per-model majority	0.54	0.68
shared probe (Section 2 design)	0.58	0.68
self probe, direct, best depth	0.61	0.57
self probe, solve rate then robustness	0.61 to 0.63	0.45 to 0.51
self probe, blend (submitted)	0.63	0.64
sampled base accuracy (8 generations, reference)	0.74	n/a

Table 1. Out-of-fold accuracy. Neighbouring layer bands and ridge strengths of the blend give 0.63 to 0.64. 117 mixed rows put one standard error near 4.5 points.

Solve rate is linearly readable. A ridge probe on the model's own last-token hidden state predicts its solve rate over 8 samples at $r = 0.69$ to 0.75 out of fold (0.59 to 0.68 for mean pooling). For comparison, the agreement of four sampled answers (fraction equal to their majority, no gold answer needed) correlates with the solve rate of four other samples at $r = 0.53$. One forward pass of a few hundred tokens beats four reasoning traces of up to 8k tokens each.

Robustness is not, once model identity is removed. Table 1 mixes two signals: which problem, and which model. To separate them we drew balanced subsets from the mixed problems, one robust and one spurious model per problem (2000 draws, 2 seeds), and scored label-free decision rules on the blend's out-of-fold scores (Figure 2).

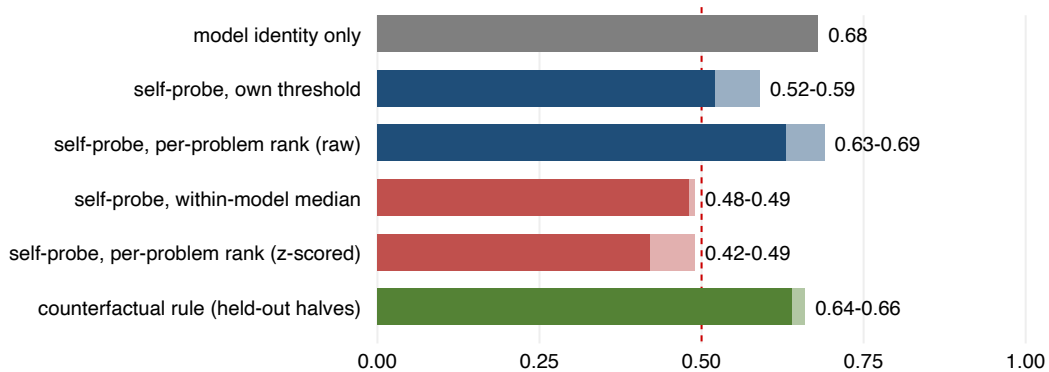


Figure 2. Accuracy on per-problem balanced draws of our labels (chance 0.50, dashed). Ranges span seeds. Rules that remove model identity (red) fall to chance; rules that keep it (blue) do no better than knowing which model is more often robust (grey). The counterfactual rule (green) is evaluated on the 38 mixed problems where variants exist, using disjoint halves of the samples for prediction and labels.

Knowing only which of the two models is more often robust already gives 0.68. Ranking the two models by the probe's raw score gives 0.63 to 0.69, no better. Ranking by scores standardized within each model, or thresholding each model at its median, gives 0.42 to 0.49. The probe's apparent advantage over the shared probe

is therefore a model-identity prior, which is exactly what the official validation and test sets are balanced to remove. On those sets we expect self-probes to be near chance, and the leaderboard agrees: the submitted self-probe also scored 0.643 (9/14), identical to the shared probe.

5. A cheap test-time counterfactual check does not transfer

Since the representation did not carry within-model robustness, we tried measuring behaviour at test time, which the rules permit within the time limit. The submission asks the target model to answer the original twice and one variant per family once, with a budget planner that shrinks the reasoning budget (8k down to 1k tokens) to fit 3600 s and backs off on out-of-memory errors; problems that do not fit fall back to the self-probe. With no gold answer, the model's majority answer on the original serves as pseudo-gold, and a problem is robust if no family loses more than 0.1 of it.

Evaluation	counterfactual	reference
our labels, balanced mixed problems (held-out halves)	0.64 to 0.66	self-probe within model 0.42 to 0.49
34 official rows, offline (RTX 5090, 3072 s)	0.38 (balanced 0.48)	always-true 0.68 (balanced 0.50)
Codabench validation, 14 cases	0.50	self-probe 0.643

Table 2. The counterfactual rule against our labels and against the organizers' labels.

Against the organizers' labels the rule calls most cases spurious (9 of 34 predicted robust, while 70% are labeled robust). With five samples per problem a single unstable answer flips the call, whereas the organizers label with about ten samples, budgets up to 100k tokens and LLM-written perturbation families (rephrase, domain, expert) that rule-based variants do not reproduce. The agreement with our own labels came from sharing those choices, not from measuring the same property.

6. Discussion

- **Check label structure before trusting a probe.** The official training rows contain no problem with different labels across models; any probe validated only there can be a difficulty detector. Evaluating on per-problem balanced subsets, and against a model-identity-only baseline, exposed this for both our shared and self probes.
- **Prompt representations encode "can I solve this?" but not "will I stay right under perturbation?".** The first is a property of model and problem that is decided before reasoning starts; the second depends on how stable the reasoning path is, which a prompt-only probe did not capture in our data.
- **Self-generated labels inherit their generator's choices.** Budget, perturbation families and sample counts determine what "robust" means. Two of our runs produced plausible-looking but meaningless labels (truncation, fp8 KV) and a third produced a rule that agreed only with itself.

Limitations. 150 problems and three models; no Olmo self-labels; 8k reasoning budget with forced answers instead of 100k; rule-based perturbations only; the never-solved-is-robust convention (without it only 56 mixed rows remain and the direct self-probe reaches 0.65 mixed / 0.61 official); the 14-case leaderboard cannot separate our entries; the official rows used for checking are few (34).

7. Reproducibility

Code, labels and per-experiment results: <https://github.com/byeolki/aimo-interp>. The final submission is `submissions/self-probe` (ZIP sha256 prefix 114d4b2a6539, built deterministically by `scripts/build.py self-probe`); its artifacts are exported by `experiments/02-self-labels/export_blend.py` from

data/self_labels.jsonl and hidden states extracted by experiments/01-layer-probe/extract.py. Labels are produced by experiments/02-self-labels/run_models.sh (vLLM 0.31.0, bf16 KV for Skywork). Probe training and inference use the evaluation image's pinned versions (PyTorch 2.12.1, Transformers 5.13.0, scikit-learn 1.8.0); probes are stored as plain NumPy arrays. The offline contract test (official ingestion, network disabled) runs in 38 s for the 34 official rows with 0 invalid predictions and identical outputs across runs. Total rented compute for all experiments was about USD 10 on RTX 5090-class GPUs.

References

[1] AIMO Interpretability Challenge. Competition proposal, NeurIPS 2026 Competition Track. arXiv:2607.13899, 2026.
Website and rules: <https://aimo-interp.github.io/>